

1 An admissibility test for constitutive claims about
2 phenomenal presence: necessary conditions, asymmetric
3 falsification, and a conjunctive return-compatible
4 criterion

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6 **Title page**

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14 **Article type.** Research Article (theoretical and methodological, with simulation-based
15 characterization)

Abstract

Correlational or causal evidence does not by itself show that a mechanism constitutes experience: a stable association, temporal precedence, or a necessary intervention are each compatible with a mechanism that causes experience without constituting it. We propose an admissibility test for constitutive claims about phenomenal presence. The test is asymmetric: a claim acquires empirical content only if it derives a non-trivial necessary physical condition, connects it to explicit observation and presence-attribution models, and accepts in advance a region of failure under a severe test. Reliable failure of a core condition can exclude the formulation; survival confirms neither constitution nor identity. We specify eight admissibility requirements, a separation between a claim’s theoretical domain and the domain a study evaluates, validity gates, and four outcomes. The worked example is a closed recurrent process in which an early source state propagates through an intermediate population and returns to it. Its observational criterion is conjunctive, requiring predictive gain along each segment, preservation of episode identity across the chain, and a preregistered guarantee that early and late windows measure the same population. In simulations of twelve ground-truth architectures the conjunction rejects five architectures lacking return in 99.5-100% of replicates, while a token-disruption architecture yields positive predictive gain on both segments as chain-level specificity reaches only 30.0%, so that component is not redundant. The conjunction still survives in every study under a trial-specific latent feedforward state without return. Sharpening an observational criterion improves discrimination without delivering identification.

Keywords: consciousness; phenomenal presence; falsification; constitution; causation; necessary condition; severe testing; equivalence testing; recurrent return; conditional independence

1. Introduction

Research on consciousness has moved from the search for correlates to the comparison of theories with explicit predictions. An inferential gap nonetheless persists: a mechanism may *correlate* with experience, *causally contribute* to it, or *constitute* it, and a stable association, temporal precedence, or a necessary intervention do not by themselves decide which relation holds (Chalmers 2000; Miller 2014; Kozuch and Kriegel 2015). The problem is sharpest when the explanandum is phenomenal presence rather than reportability, because presence has no index independent of access.

1.1 Scope

What follows offers an admissibility criterion for falsificationist tests of strong constitutive claims: it specifies what a theory must declare so that a result can exclude its current formulation. It does not exhaust the possible forms of constitutive evidence. It does not cover indirect evidence obtainable through mutual manipulability and its refinements (Craver 2007; Baumgartner and Gebharder 2016), inference to the best explanation, or comparative parsimony. Neither is it a criterion of scientific demarcation: a claim that fails this test is not thereby unscientific, but is not empirically evaluable by that body of data on its own.

1.2 Relation to previous work

Doerig et al. (2019) and Doerig, Schurger and Herzog (2020) proposed criteria for empirical theories of consciousness and showed that causal-structure theories face a testability problem when confronted with functionally equivalent systems. Hanson and Walker (2021) formalized falsification across a computational hierarchy, arguing that a theory must be invariant with respect to changes that leave the inference procedure fixed at a given level, and showing that a theory can be simultaneously falsified at one level and unfalsifiable at another. Kleiner and Hoel (2021) formalized the conflict between predicted and inferred experience.

Craver (2007) and Baumgartner and Gebharder (2016) analysed the mutual-manipulability criterion for constitutive relevance and its limits under interventionism (Woodward 2003). Miller (2007, 2014) and Fink (2016) developed the correlation-versus-constitution problem for the neural correlate of consciousness, and a large collaboration argued that a theory is scientifically inadmissible when its central claims are untestable even in principle (IIT-Concerned et al. 2025). The adversarial collaboration between rival theories showed that divergent predictions and pass-or-fail criteria can be preregistered (Cogitate Consortium et al. 2025).

The difference from the closest antecedent is worth stating precisely. Hanson and Walker analyse at which computational level a theory can be assessed and what invariance it must satisfy to be falsifiable at all. The present article addresses a question downstream of theirs: given a theory that does make a physical commitment, what must it declare so that a particular study can exclude it, and what may be concluded when it does not. Our units are necessary conditions, observation and presence-attribution models, revision policy, and a four-way decision.

None of the components used here is new in isolation: neither the logic of necessary conditions, nor hypothetico-deductive falsification, nor severe testing, nor the constitution-causation distinction, nor observation models, nor the common-input problem, nor out-of-sample prediction, nor conditional randomization and residual permutation. The contribution is their joint, preregistrable architecture, together with the demonstration that even a fully conjunctive observational criterion for a topological claim does not deliver identification.

Previous work	Problem addressed	What it does not provide	What this article adds
Doerig et al. (2019, 2020)	Testability criteria; functional equivalence	Criteria specific to the constitutive claim and its confirmation gap	A rule and requirements centred on that claim

Previous work	Problem addressed	What it does not provide	What this article adds
Hanson and Walker (2021)	Falsifiability across computational levels	A local decision rule for a given study	Validity gates, four outcomes, and revision policy
Kleiner and Hoel (2021)	Conflict between predicted and inferred experience	What the claim must specify in order to be able to fail	Operates one level earlier: its admissibility
Craver (2007); Baumgartner and Gebharder (2016)	Mutual manipulability and its limits	A falsificationist procedure for the case of presence	Imports the diagnosis and instantiates it operationally
Miller (2007, 2014); Fink (2016)	Correlation versus constitution for the NCC	Explicit requirements and a decision rule	Non-triviality, severity, scope, four outcomes
Cogitate Consortium et al. (2025)	Adversarial preregistration	An interpretive rule for constitutive claims	Asymmetry and prediction centrality

Table 1. Position of the framework relative to its direct antecedents.

2. Methods and conceptual framework

2.1 Explanatory relations and notation

Let M be a candidate mechanism and P phenomenal presence. An identity claim holds that $M = P$: one fact under two descriptions. A constitution claim holds that M is part of what P consists in, without requiring full identity. A realization claim holds that M realizes

91 P , possibly among alternative realizers. A causal dependence claim holds that $M \rightarrow P$.
92 This article addresses strong constitutive claims, with identity as a limiting case, and uses
93 $=$ for identity throughout. An interpretive rule governs the text: surviving the test confirms
94 neither a constitutive relation nor, a fortiori, an identity claim.

95 By **phenomenal presence** we mean the existence of phenomenal experience, abstracting
96 from its particular content and from its availability for report. The construct has affinities
97 with phenomenal as opposed to access consciousness (Block 2001), with state consciousness,
98 and with minimal phenomenal experience (Metzinger 2020); it is not asserted to be coexten-
99 sive with any of them. Metzinger’s construct is tied to a specific class of meditative states,
100 and Windt’s (2015) analysis of dreamless sleep concerns a particular sleep phenomenology;
101 both are narrower in their evidential base than the present usage, which is deliberately
102 abstract and makes no commitment about which states instantiate it.

Symbol	Meaning
T	Constitutive claim under evaluation
P, M	Phenomenal presence; candidate physical mechanism
$N_T(M; x, t)$	Necessary physical condition derived from T , for system x at time t
E_P	Evidence of presence, inferentially independent of M
D, D_{eval}	Theoretical domain; domain evaluable in the present study
$\mathcal{C}, O, \varphi, \mathcal{R}$	Experimental conditions; observation model; presence-attribution model; decision rule
$\delta_{\text{min}}, \delta_{\text{epi}}$	Smallest relevant effect for the R^2 -scale and episodic components

Symbol	Meaning
A_1, A_2, B	Source population early and late; intermediate population
X, U	Observed and unobserved shared covariates

Table 2. Notation used uniformly throughout.

2.2 A working taxonomy

Three variables must be kept apart: **presence** (P); **access**, the functional availability of information for report, reasoning, and behavioural control (Block 2001; Lamme 2003); and **arousal**, physiological activation, which modulates experience but must not be identified with presence by default. Replacing the conscious/unconscious binary with a multidimensional state space is well established (Bayne, Hohwy and Owen 2016).

Absence of report is not absence of experience. Report requires perception, memory, selection, and motor execution; its absence may reflect loss of access, motor deficit, or noise (Timmermans and Cleeremans 2015). This motivates no-report paradigms, which seek to dissociate neural activity associated with conscious experience from activity required for explicit decision and report. Tsuchiya et al. (2015) argued that report-dependent contrasts may overestimate candidate correlates by incorporating report-related processes while underestimating experience by excluding aspects difficult to communicate. No-report paradigms are an important methodological response to the access–presence problem; they are not direct detectors of experience and do not remove the need for an independently justified attribution model.

2.3 The asymmetric rule and the contradiction pattern

To acquire empirical content, T must imply at least one necessary physical condition $N_T(M; x, t)$ whose failure can exclude it, while its satisfaction does not imply the constitutive relation. Multiple causal, realizational, and constitutive claims may share the same condition, so satisfying it does not discriminate among them.

The naive form “not- N implies not- T ” is too weak: the mere absence of N contradicts nothing if there was no experience in that episode. If T asserts, for the domain D , that $P(x, t) \rightarrow N_T(M; x, t)$, the contradicting pattern is the co-occurrence, within D , of evidence of presence and reliable absence of the condition:

$$x \in D, \quad E_P(x, t) \wedge \neg N_T(M; x, t).$$

Excluding a formulation requires seven conditions simultaneously: (1) the case belongs to D ; (2) valid evidence of presence exists; (3) the condition is reliably absent; (4) the measurement has sufficient sensitivity; (5) relevant auxiliary conditions are satisfied; (6) the derivation from T is valid; and (7) the condition is core rather than auxiliary.

2.4 The theoretical domain and the evaluable domain

A recurring error is to let evidential limitations silently redefine the scope of a theory. Two objects must therefore be separated. D is the **theoretical domain**: the class of systems and states for which T asserts its condition. D_{eval} is the **evaluable domain** of a particular study:

$$D_{\text{eval}} = \{x : x \in D\} \cap \{\varphi(Z, S) \geq \tau_P\} \cap \{\text{all validity gates passed}\}.$$

An episode may satisfy $x \in D$ while $x \notin D_{\text{eval}}$. The correct description is then that the

episode lies within the theoretical domain but outside the evaluable domain of the present analysis, and is therefore not eligible for contradiction in that dataset. It is not a narrowing of D . A theory whose D contracts after each unfavourable dataset is protecting itself by the mechanism described in §2.7, and the contraction must be recorded as a substantive modification rather than as a clarification.

2.5 Eight admissibility requirements

A condition is admissible when it simultaneously satisfies **derivation**, **non-triviality**, **discrimination**, **operationalization**, **severity** (Mayo 2018), **inferential independence**, **theoretical stability**, and **scope specificity**. Their joint function is to prevent a theory from resting on conditions that are trivial, circular, chosen post hoc, unmeasurable, immunized against failure, or lacking an explicit domain. No historical originality is claimed for any requirement separately.

Requirement	Control question	Inadequate example	Consequence of failure
Derivation	Does it follow from the theory's core?	Condition chosen after inspecting the data	Post hoc adjustment, not a test
Non-triviality	Is there any system that violates it?	"Requires physical activity"	The theory cannot lose
Discrimination	Does it exclude some class of alternatives?	Condition shared by all rivals	Adjudicates nothing
Operationalization	Are observable and procedure defined?	"Sufficient integration" without a measurement function	Not evaluable

Requirement	Control question	Inadequate example	Consequence of failure
Severity	Would its absence be detected if it truly failed?	Low-powered design	Survival without informative content
Inferential independence	Is it defined without the outcome it explains?	Biomarker that defines presence	Circularity
Theoretical stability	Does abandoning it change the theory?	A new condition after each failure	Indefinite escape
Scope specificity	For which states, species, and scales?	Condition without a domain	Indeterminate scope

Table 3. Requirements that render a necessary condition admissible.

“Consciousness requires physical activity” is inadmissible by triviality; “consciousness requires some neural interaction” is weakly informative; “in conscious visual experience under conditions $\mathcal{C}$, activity must return to source population A through intermediate population B within window W , preserving episode-specific information, after controlling feedforward activity, baseline state, and common input” is potentially admissible.

2.6 Relative admissibility, multiple conditions, and centrality

A theory is admissible only relative to a condition, conditions, domain, observation model, attribution model, and decision rule:

$$A(T \mid N, \mathcal{C}, D, O, \varphi, \mathcal{R}).$$

In the conjunctive case, $T \rightarrow N_1 \wedge N_2 \wedge \dots$, reliable failure of any genuinely necessary

condition excludes the formulation, but survival of one does not entail survival of the others.
The worked example of §3 is conjunctive, and §3.2 states the rule that governs it.

Disjunctive conditions. If T implies $N_1 \vee N_2$, the absence of N_1 alone does not contradict T : the counterexample requires E_P together with the joint absence of all admissible disjuncts. For the disjunction not to render the theory irrefutable, it must be declared before the result, be finite and exhaustively enumerated, have each disjunct separately satisfy the eight requirements, and fall under the revision rule of §2.7, so that adding a disjunct after a failure counts as a substantive modification.

Centrality. Core predictions, whose failure requires abandoning the theory; **implementation-level** predictions, revisable without abandoning the abstract principle; and **auxiliary** predictions, contingent on experimental assumptions. Only failure of a core condition satisfies requirement (7) of §2.3.

2.7 Observation and presence-attribution models, and the revision rule

The observation model O specifies how measurements relate to the condition. Formally one observes $A^* = A + \epsilon_A$ and $B^* = B + \epsilon_B$, and O must address measurement error, reliability, temporal jitter, spatial mixing, autocorrelation, leakage between windows, insufficient resolution, and latent variables.

The presence-attribution model is a function $\varphi(Z, S) = \Pr(P = 1 \mid Z, S)$, where Z collects reports, behaviour, ocular, autonomic, or other evidence and S is the state class, together with two prespecified thresholds: $\varphi \geq \tau_P$ indicates sufficient evidence of presence, $\varphi \leq \tau_A$ insufficient, and intermediate values indeterminate.

φ does not measure experience. It is a declared degree of inferential support, calibrated on states where the evidence is least contested and extrapolated, with acknowledged error,

185 to states where it is not. Three independences must be distinguished. **Statistical inde-**
186 **pendence** concerns whether Z and M are uncorrelated, which they typically are not and
187 need not be. **Measurement independence** concerns whether Z is recorded through a
188 channel distinct from the one measuring M . **Inferential independence** concerns whether
189 the justification for attributing presence appeals to M itself. Only the third is mandatory:
190 attributing presence by means of the very property under evaluation is circular and voids
191 requirement (2) of §2.3. The framework does not solve the consciousness-measurement prob-
192 lem; it requires that the inferential bridge be declared and audited.

State class S	Evidence Z	Calibration	Dominant errors	Rule
Wakefulness, reliable report	Concurrent report	Reference class; φ near its ceiling	Post-perceptual cognition; criterion bias	$\varphi \geq \tau_P$: suffi- cient
Wakefulness, no report	Ocular and autonomic markers	Calibrated against the same task with report	Sensitivity to unconscious processing; drift	Typically $\varphi \geq \tau_P$; declare drift
Light sedation	Delayed report; behaviour	Extrapolated from wakefulness	Amnesia; access–presence dissociation	Often inde- termi- nate
Deep NREM sleep	Report after awakening	Weak; asymmetric errors	False negatives from amnesia	Usually inde- termi- nate

State class S	Evidence Z	Calibration	Dominant errors	Rule
No behaviour, no report	No positive evidence	Not calibrated	Absence of evidence is not evidence of absence	Within D but outside D_{eval}

Table 4. Operational specification of φ . Thresholds are declared before analysis. No row licenses attributing presence by means of M . The final row records an evidential limitation of the study, not a restriction of the theory’s domain.

Theory continuity. Before testing, the admissible response to each result must be fixed. A **refinement** sharpens scope or operationalization without altering content. An **implementation change** replaces a concrete realization while leaving the abstract principle intact; the adverse result is recorded as a failure of that implementation, not as confirmation of the core. A **core replacement** substitutes a core condition after a failure: a change that breaks explanatory continuity and obliges the new formulation to be assessed as a distinct claim with its own preregistration. A **successor theory** arises when the resulting formulation no longer shares the original constitutive commitment; it does not inherit the survival accumulated by its predecessor. A theory replacing its necessary condition after every failure avoids refutation only at the cost of explanatory continuity.

2.8 Hierarchical validity gates and four outcomes

The decision rule $\mathcal{R}$ applies validity gates in a fixed order *before* any component test: (1) domain membership; (2) valid evidence of presence under φ ; (3) adequate observation model; (4) sufficient reliability; (5) sufficient sensitivity; (6) positive control passed; (7) minimal estimability and design adequacy. Failure of any gate yields **not evaluable**.

The seventh gate is deliberately named for what it establishes. An observations-per-

parameter criterion, a conditioning number, or a fold-to-fold stability check speaks only to whether the model is numerically and statistically estimable with minimal stability. It says nothing about causal or statistical identification of the target relation. Causal identification belongs to Level 2 of §3.6 and requires intervention, exogenous variation, or explicit and defensible causal assumptions.

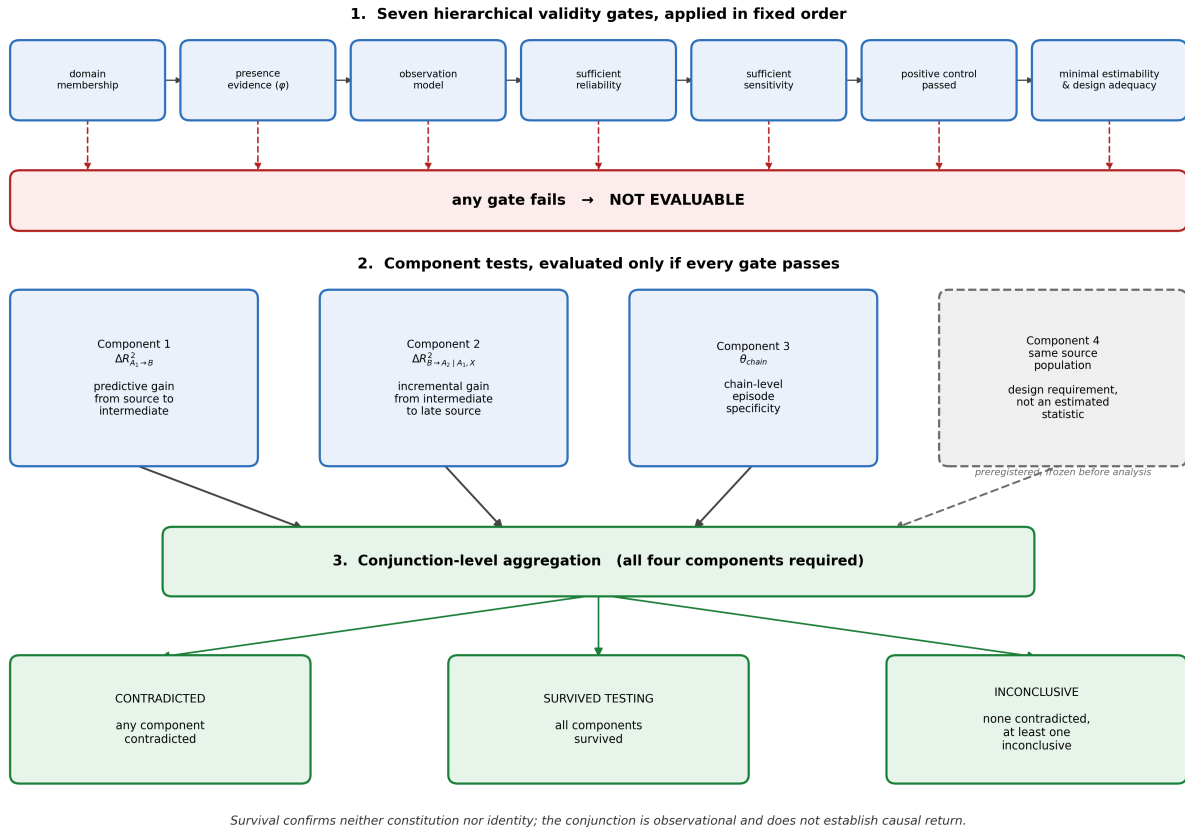


Figure 1: Structure of the decision procedure. Seven validity gates are applied in fixed order; failure of any gate yields “not evaluable”. Only if every gate passes are the four components of the conjunctive criterion evaluated, three of them estimated from data and the fourth a preregistered design requirement. The conjunction is then aggregated into one of three outcomes.

Alt text: A three-stage flow diagram. The top stage shows seven boxes in a row, labelled domain membership, presence evidence, observation model, sufficient reliability, sufficient sensitivity, positive control passed, and minimal estimability and design adequacy; each has a dashed red arrow down to a wide red bar reading “any gate fails, not evaluable”. The middle

stage shows four component boxes: predictive gain from source to intermediate, incremental gain from intermediate to late source, chain-level episode specificity, and a grey dashed box marked same source population, design requirement, not an estimated statistic. The bottom stage shows a green conjunction-level aggregation box branching to three outcome boxes: contradicted, survived testing, and inconclusive.

The four outcomes are: **contradicted**, adequate evidence of the absence of the condition within D_{eval} with the complete pattern of §2.3; **survived testing**, evidence compatible with the condition, confirming neither constitution nor identity; **inconclusive**, a valid design whose uncertainty remains too wide to decide; and **not evaluable**, a failure of validity, measurement, sensitivity, or estimability. Measurement error is treated in two ways: error that is high but quantified, with the components still estimable, can yield inconclusive; error that prevents estimation, or reliability below the declared threshold, yields not evaluable.

Outcome	Condition	Theoretical consequence
Contradicted	Valid evidence of absence in D_{eval} , complete pattern of §2.3	Excludes the formulation if the condition is core
Survived testing	Evidence compatible with the condition	Retains the theory as a candidate; confirms nothing
Inconclusive	Valid design, uncertainty too wide	None; greater power required
Not evaluable	Failure of a validity gate	None; a different design required

Table 5. The four outcomes and their consequences.

3. Worked example: a conjunctive return-compatible criterion

3.1 The claim

Recurrence is used as a worked case, not as a theory endorsed here. The example matters because a simple early-late dependence, written $E \rightarrow L$, is *not* recurrence: it is equally consistent with local persistence, unidirectional feedforward propagation, or a shared latent cause. Recurrence requires return to the source. The example claim is:

T_R : within the declared domain D , phenomenal presence requires a closed recurrent process in which an early state of a source population A propagates through an intermediate population B and subsequently returns to A , preserving episode-specific information and altering its later state.

The claim is about informational return and the alteration of a later state. It does not assert a state-dependent interaction between the returning signal and the population it re-enters, and no analysis reported here would test such an assertion.

3.2 The conjunctive observational criterion

From T_R we derive an observational criterion that is conjunctive by construction:

$$N_R^{\text{obs}} = N_{A_1 \rightarrow B} \wedge N_{B \rightarrow A_2 | A_1, X} \wedge N_{\text{chain}} \wedge N_{\text{same-source}}.$$

$N_{A_1 \rightarrow B}$. The early source window provides out-of-sample predictive information about the intermediate population beyond observed covariates:

$$\Delta R_{A_1 \rightarrow B}^2 = R^2(B \sim X + A_1) - R^2(B \sim X).$$

252 $N_{B \rightarrow A_2 | A_1, X}$. The intermediate population provides out-of-sample predictive information
 253 about the late source window beyond the early window and observed covariates:

$$\Delta R_{B \rightarrow A_2}^2 = R^2(A_2 \sim X + A_1 + B) - R^2(A_2 \sim X + A_1).$$

254 This is the **intermediate-population incremental estimand**. On its own it is silent about
 255 whether anything reached B from A_1 , and it is therefore not a return-specific quantity.

256 N_{chain} . Episode-specific information is preserved across the full chain $A_{1,i} \rightarrow B_i \rightarrow A_{2,i}$,
 257 defined in §3.4.

258 $N_{\text{same-source}}$. A_1 and A_2 are measurements of the same preregistered source population. This
 259 is a design requirement, not a quantity estimated from data; §3.3 states how it is discharged.

260 Only the complete conjunction may be described as **compatible with recurrent return**.
 261 No single component, and no proper subset, is evidence of recurrent return. Even the
 262 complete conjunction is observational: it does not establish causal return, and §4.1 shows
 263 that it can be satisfied in full when no return exists.

264 N_R^{obs} must be distinguished from local persistence of A , autocorrelation, temporal smearing,
 265 leakage between windows, feedforward routes terminating elsewhere, baseline activity, stimu-
 266 lus, arousal, observed common causes, latent common causes, trial-specific latent feedforward
 267 states, and shared low-rank factors.

268 **Decision rule.** If any validity gate fails, the study is not evaluable and no component is
 269 assessed. Otherwise: if any required component is contradicted under a valid design, the
 270 conjunction is **contradicted**; if all required components survive testing, the conjunction
 271 **survived testing**; if none is contradicted but at least one is inconclusive, the conjunction is
 272 **inconclusive**. The conjunction is never classified as survived when only some components
 273 survive.

On temporal precedence. Precedence does not simply eliminate the reverse direction. Under adequate alignment and measurement assumptions it rules out direct retroactive causation between the measured variables, but it does not eliminate temporal smearing, non-causal filtering, source mixing, jitter, or underlying bidirectional dynamics sampled at two times.

3.3 The same-source-population requirement

The claim is about return to a source, so the identity of that source across windows is not a detail. Three notions must be distinguished and are not automatically equivalent: the **same anatomical population**, the **same functional population**, and the **same recording channels**. Channels may be stable while the functional population they sample drifts; a functional population may be stable while the channels sampling it change across sessions.

The worked example requires the same preregistered recording units, mapped to the same anatomical population. Acceptable preregistered specifications include the same electrodes, intracranial contacts, sorted units, vertices, voxels, anatomically registered region, predefined channel set, or a mapping learned exclusively from training data and frozen before the test partition is touched.

If late activity is measured in a population other than the one measured early, a positive result may indicate sequential propagation but cannot indicate return to the original source. Post hoc relocation of A after inspecting results constitutes a core replacement under §2.7 rather than a refinement.

3.4 Chain-level episode specificity

The chain component comprises two linked matching problems.

Link 1. Fit $B \sim X$ and $B \sim X + A_1$ on the training partition. On held-out trials the query is $q_i = \widehat{B}_i^{X+A_1} - \widehat{B}_i^X$ and the target is $r_i = B_i - \widehat{B}_i^X$.

298 **Link 2.** Fit $A_2 \sim X + A_1$ and $A_2 \sim X + A_1 + B$ on the training partition. The query is
 299 $q_i = \widehat{A_{2,i}}^{X+A_1+B} - \widehat{A_{2,i}}^{X+A_1}$ and the target is $r_i = A_{2,i} - \widehat{A_{2,i}}^{X+A_1}$.

300 **Matching.** Held-out trials are partitioned into strata $s \in \mathcal{S}$ defined by stimulus category
 301 crossed with arousal quartile; strata with fewer than three trials are discarded. Within each
 302 stratum, each query selects its nearest target under squared Euclidean distance, **independ-**
 303 **ently of every other query:**

$$\hat{j}(i) = \arg \min_{j \in \mathcal{S}} \|q_i - r_j\|^2.$$

304 Matching is therefore independent nearest-neighbour assignment, not a one-to-one assign-
 305 ment: **a target may be selected by more than one query**, no query ordering is imposed,
 306 and the result does not depend on the order in which trials are processed. Ties are resolved
 307 by the lowest index, which occurs with probability zero for continuous data. Accuracy and
 308 chance for a link are

$$a = \frac{1}{n_{\text{te}}} \sum_{s \in \mathcal{S}} \sum_{i \in s} \mathbf{1}[\hat{j}(i) = i], \quad c = \frac{|\mathcal{S}|}{n_{\text{te}}},$$

309 where n_{te} is the number of retained held-out trials. The chance term follows because, under
 310 independent matching within a stratum of size m_s , each query selects its own target with
 311 probability $1/m_s$, and $\sum_s m_s \cdot (1/m_s) = |\mathcal{S}|$. Chance is computed rather than assumed.

312 **Link and chain effects.** For link $k \in \{1, 2\}$ the effect is the accuracy above chance,

$$\theta_{\text{link } k} = a_k - c_k,$$

313 and the chain effect is the weakest link,

$$\theta_{\text{chain}} = \min(\theta_{\text{link } 1}, \theta_{\text{link } 2}).$$

The minimum is the natural aggregation for a conjunctive requirement: the chain preserves episode identity only to the extent that its least informative segment does, and a product or mean would allow a strong link to compensate for a broken one, which is precisely the failure mode the component exists to detect.

Two decision procedures use these quantities, and they differ. In the **architecture characterization** (§4.1), each link is tested separately against a null generated by permuting targets within strata while holding queries fixed, with 100 permutations per link; the component is positive only if both links reach $p < 0.05$. In the **four-outcome classification** (§4.2–§4.4), a single equivalence decision is taken on θ_{chain} against the margin δ_{epi} , using a bootstrap interval over subjects, exactly as for the two R^2 -scale components. Both are conjunctive in effect; they are reported separately because they are not the same procedure.

All model fitting, residualization, and any hyperparameter choice occur inside the training partition only. No dimensionality reduction, scaling, or metric learning is applied, so no such transformation can leak.

3.5 Conditional nulls

Each segment estimand has its own null hypothesis, and the two are not interchangeable.

First segment. $H_0^{A_1 \rightarrow B} : A_1 \perp\!\!\!\perp B \mid X$. The null is generated by residual permutation. A_1 is regressed on X using the training partition only; the training-fitted coefficients are applied to all trials to form fitted values and residuals; residuals are permuted and added back to the fitted values; the statistic is evaluated on the held-out partition.

Second segment. $H_0^{B \rightarrow A_2} : B \perp\!\!\!\perp A_2 \mid A_1, X$. The procedure is identical with the nuisance block $[X, A_1]$.

The nuisance block for the second segment must include A_1 . Permuting B within strata of stimulus category and arousal alone would destroy the legitimate association between A_1 and B and so test a null that no architecture of interest satisfies. The residual scheme preserves $P(B \mid A_1, X)$ as estimated by the conditional model, which is the null actually required.

Exchangeability and its limits. The scheme assumes that, under the null, residuals of the permuted variable with respect to the nuisance block are exchangeable across trials. This holds exactly only when the conditional model is correctly specified. Under misspecification the null is approximate and the test may be anticonservative; no claim of exactness is made. A conditional randomization test drawing surrogates from an estimated conditional distribution (Candès et al. 2018) is preferable for empirical application, where the covariate structure is not known by construction, and the misspecified-covariate scenario of §4.2 is included precisely to expose this failure mode.

3.6 Three inferential levels

Level 1, compatible with recurrent return. The full conjunction N_R^{obs} is satisfied. This level remains vulnerable to latent confounding, persistence, and misspecification, as §4.1 demonstrates quantitatively.

Level 2, causally supported recurrent return. The return path receives support through selective perturbation of B , selective perturbation of the $B \rightarrow A$ pathway, exogenous variation, or explicit and defensible identification assumptions. The intervention must reasonably preserve stimulus, arousal, the initial feedforward sweep, global signal quality, and early activity in A .

Level 3, necessary constitutive condition survived testing. Even a demonstrated causal return shows only that the necessary condition survived. It does not show that recurrence constitutes presence or is identical to it.

3.7 Statistical specification

Out-of-sample R^2 is computed as $1 - \text{SS}_{\text{res}}/\text{SS}_{\text{tot}}$ on a held-out partition, with the reference being the training-set mean of the multivariate target; sums are taken over all target dimensions jointly, so a single scalar summarizes the multivariate fit. Negative values are retained rather than truncated, since truncation biases the difference of two R^2 values. Trials are split 50/50 into training and test partitions within subject. The regularization parameter is fixed at unity in the reported simulations, a value validated by the recovery results of §4 rather than tuned on the data; in an empirical application it should be selected by nested internal validation within the training folds, and grouping by session, run, or block should replace random splitting so that temporally adjacent trials do not straddle the partition boundary. Effects are estimated within subject and aggregated across subjects by bootstrap resampling of subjects, with 90% intervals used for the two-sided equivalence decision. A component is deemed **survived** when the interval lies entirely above δ ; **contradicted** when the interval lies within $\pm\delta$, and also when it lies entirely below $-\delta$, since an effect reliably in the adverse direction is incompatible with a component that asserts a positive predictive gain; and **inconclusive** otherwise. The adverse-direction case did not arise in any of the 1200 component decisions across the twelve scenarios reported here, so no reported classification depends on it; it is stated because it can arise in empirical application. The margins are $\delta_{\text{min}} = 0.01$ for the two R^2 -scale components and $\delta_{\text{epi}} = 0.02$ for θ_{chain} . Both are illustrative of the generative model rather than empirically calibrated; §4.4 reports sensitivity to that choice. In an application they should be derived from test–retest reliability, positive-control performance, artefact magnitude, instrumental precision, the theoretical consequences of an effect of that size, and simulations calibrated to the paradigm.

3.8 Simulation design

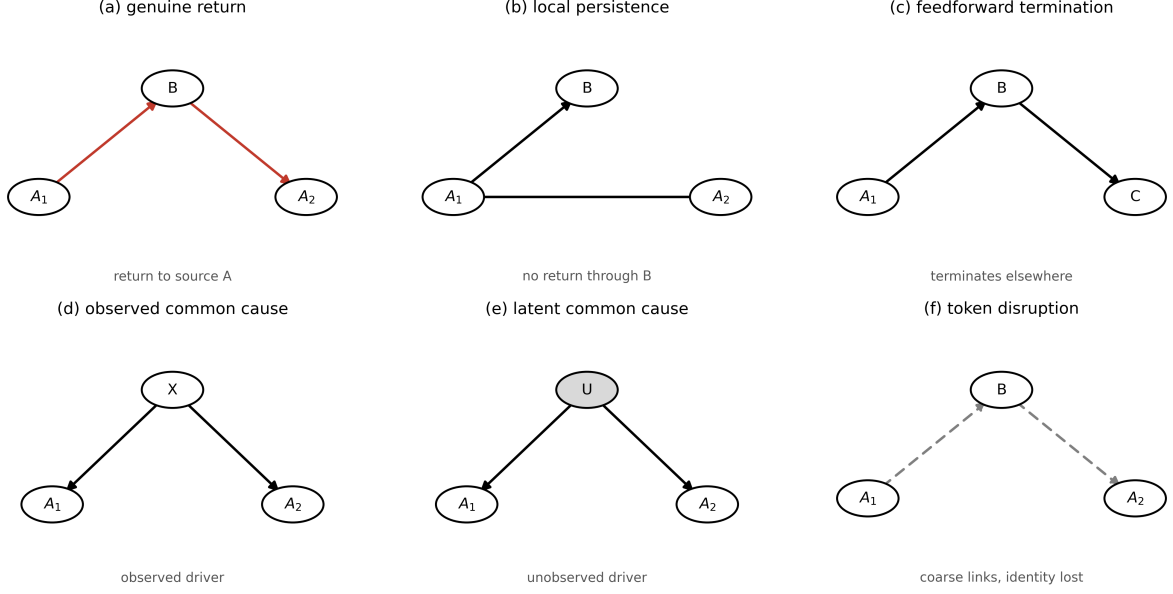
Twelve architectures were generated with known ground truth, where the truth flag records whether A_2 receives episode-specific return through B : **recurrent return**; **nonlinear return**; **latent confounder plus weak return**; **local persistence**, in which $A_1 \rightarrow A_2$ directly with B a side branch driven by A_1 ; **feedforward chain**, terminating outside A ; **observed common cause**; **parallel pathways**; **latent confounder**; **latent trial-specific feedforward state**; **shared low-rank factor**; **coarse dependence**; and **token disruption**.

The token-disruption architecture tests whether the chain component is redundant with the two predictive-gain components. In it, A_1 drives B only through a five-level coarse summary of A_1 , and B drives A_2 only through a five-level coarse summary of B . Both predictive gains are therefore genuinely positive, while the episode-specific token originating in A_1 is destroyed at each link. If the chain component added nothing, this architecture would satisfy the conjunction.

Each replicate used 1200 trials with six-dimensional populations, four stimulus categories, a standard normal arousal covariate, and unit-scale Gaussian noise; 200 replicates per architecture.

Alt text: Six small directed graphs. Panel a shows A_1 with a red arrow to B and a red arrow from B to A_2 , labelled return to source A . Panel b shows A_1 with arrows to both B and A_2 , labelled no return through B . Panel c shows A_1 to B to C , labelled terminates elsewhere. Panel d shows an observed node X with arrows to A_1 and A_2 . Panel e shows the same with a grey unobserved node U . Panel f shows A_1 to B to A_2 with both arrows dashed and grey, labelled coarse links, identity lost.

For the four-outcome classification, twelve scenarios were generated and 200 studies simulated per scenario. A study comprised 20 subjects with 200 trials each, except the six-subject scenario and a high-power scenario using 40 subjects with 400 trials. Complete generative



Only panel (a) contains episode-specific return to the source population. A_1 and A_2 denote the same population in an early and a late window; B is the intermediate population. Red marks the return path; grey fill marks an unobserved variable; dashed grey marks a link that transmits only a coarse summary, so episode identity is not preserved.

Figure 2: Six architectures that can generate segment-wise predictive dependence. Only (a) contains episode-specific return to the source population; panel (f) transmits only a coarse summary along each link, so both predictive gains are positive while episode identity is destroyed in transit.

equations, coefficients, seeds, and the measured effect of every scenario are given in the Supplementary Methods.

4. Results

4.1 Separation of causal architectures

Architecture	Return?	$\Delta R^2_{A_1 \rightarrow B}$	$\Delta R^2_{B \rightarrow A_2}$	Chain	Conjunction
Recurrent return	Yes	100.0 [98.1, 100]	100.0 [98.1, 100]	100.0 [98.1, 100]	100.0 [98.1, 100]
Nonlinear return	Yes	100.0 [98.1, 100]	100.0 [98.1, 100]	99.5 [97.2, 99.9]	99.5 [97.2, 99.9]

Architecture	Return?	$\Delta R_{A_1 \rightarrow B}^2$	$\Delta R_{B \rightarrow A_2}^2$	Chain	Conjunction
Latent confounder + weak return	Yes	100.0 [98.1, 100]	100.0 [98.1, 100]	99.0 [96.4, 99.7]	99.0 [96.4, 99.7]
Local persistence	No	100.0 [98.1, 100]	4.0 [2.0, 7.7]	2.0 [0.8, 5.0]	0.0 [0.0, 1.9]
Feedforward chain	No	100.0 [98.1, 100]	3.5 [1.7, 7.0]	6.5 [3.8, 10.8]	0.5 [0.1, 2.8]
Observed common cause	No	4.5 [2.4, 8.3]	3.0 [1.4, 6.4]	0.0 [0.0, 1.9]	0.0 [0.0, 1.9]
Parallel pathways	No	4.5 [2.4, 8.3]	4.5 [2.4, 8.3]	0.0 [0.0, 1.9]	0.0 [0.0, 1.9]
Latent confounder	No	100.0 [98.1, 100]	100.0 [98.1, 100]	86.5 [81.1, 90.6]	86.5 [81.1, 90.6]
Latent feedforward state	No	100.0 [98.1, 100]	100.0 [98.1, 100]	100.0 [98.1, 100]	100.0 [98.1, 100]
Shared low-rank factor	No	100.0 [98.1, 100]	100.0 [98.1, 100]	49.0 [42.2, 55.9]	49.0 [42.2, 55.9]
Coarse dependence	No	4.5 [2.4, 8.3]	7.5 [4.6, 12.0]	0.0 [0.0, 1.9]	0.0 [0.0, 1.9]
Token disruption	No	97.5 [94.3, 98.9]	99.5 [97.2, 99.9]	30.0 [24.1, 36.7]	30.0 [24.1, 36.7]

Table 6. Positive rates by architecture over 200 replicates, with 95% Wilson intervals. The conjunction requires all three estimated components jointly; the same-source component is a design requirement and is satisfied by construction in every simulated architecture.

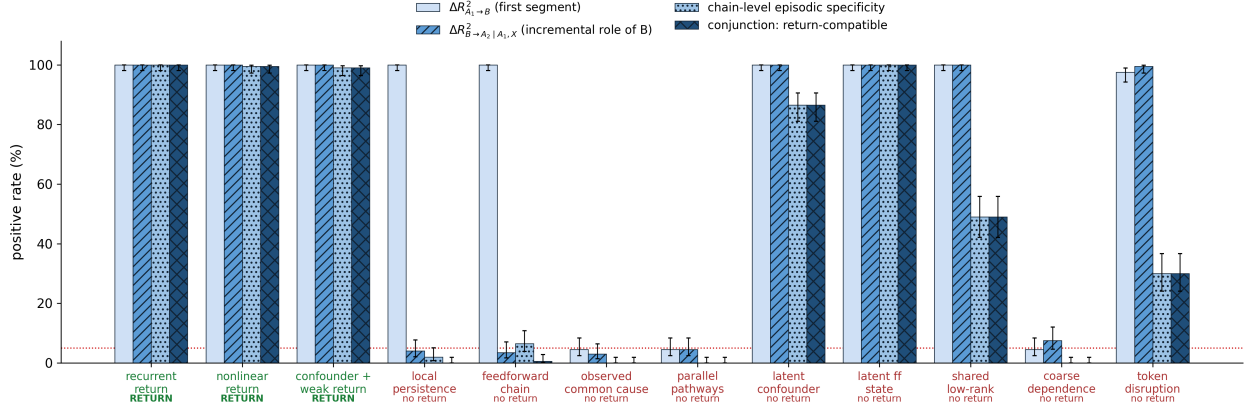


Figure 3: Positive rates for the three estimated components and for the conjunction across twelve architectures, with 95% Wilson intervals. The dotted line marks the nominal 5% level. Labels beneath the axis mark which architectures contain genuine return.

Alt text: Grouped bar chart with twelve architectures on the horizontal axis and positive rate on the vertical axis. Four bars per architecture, distinguished by fill and hatching, with error bars. The three return architectures show all four bars near one hundred. Local persistence, feedforward chain, observed common cause, parallel pathways and coarse dependence show a conjunction bar at or near zero. Latent feedforward state shows all four bars at one hundred despite containing no return. Token disruption shows the first two bars near one hundred and the last two near thirty.

Five findings follow.

First, **the conjunction discriminates far better than any single component.** All three architectures containing genuine return satisfy it in 99.0% of replicates or more. Five of the nine architectures without return yield a positive conjunctive result in only 0.0–0.5% of replicates — that is, they are rejected in 99.5–100% of replicates: local persistence, feedforward chain, observed common cause, parallel pathways, and coarse dependence. A criterion resting on early–late dependence alone would accept local persistence in every replicate and

coarse dependence in almost every replicate.

Second, **the first-segment component carries information that no other component supplies.** Local persistence and the feedforward chain both yield $\Delta R_{A_1 \rightarrow B}^2$ positive in every replicate, correctly, because in both architectures B genuinely receives from A_1 ; they are excluded by the second segment. Observed common cause, parallel pathways, and coarse dependence are excluded at the first segment instead. Without the first-segment estimand these two modes of failure would be indistinguishable, and an architecture in which nothing reaches B could pass on the strength of the second segment alone.

Third, **the chain component is not redundant.** The token-disruption architecture yields positive predictive gains on both segments, 97.5% and 99.5%, while chain-level specificity reaches only 30.0% [24.1, 36.7]. An analysis using the two predictive-gain estimands alone would have classified this architecture as return-compatible in nearly every replicate. The chain component also reduces the shared low-rank factor from full acceptance on both gain components to 49.0%.

Fourth, **latent structure defeats the complete conjunction.** The trial-specific latent feedforward state satisfies all components in 100% [98.1, 100] of replicates with no return present, and the latent confounder in 86.5% [81.1, 90.6]. A positive conjunctive result in an individual study does not identify whether a weak genuine return is present in addition to latent confounding. The components are not mistaken about the dependence, which genuinely exists and is correctly detected; the inferential error would lie in reading that dependence as recurrent return. Requiring the full chain sharpens discrimination among architectures whose structure is observable without resolving identification when the shared driver is unobserved.

Fifth, **the components fail in interpretable ways rather than uniformly**, which is what makes the conjunction diagnostic rather than merely conservative: the pattern of component failures narrows the class of compatible alternative architectures, without singling

456 out one generative model.

457 **4.2 Four-outcome classification of the conjunction**

Scenario	Contradicted	Survived	Inconclusive	Not evaluable
Absent return, adequate sample	100.0	0.0	0.0	0.0
Absent return, reduced sample (6 subjects)	72.5	0.0	27.5	0.0
Effect within the equivalence zone	95.0	0.0	5.0	0.0
Effect within the zone, larger sample ($40 \times$ 400)	100.0	0.0	0.0	0.0
Effect near the boundary	4.0	9.5	86.5	0.0
Effect clearly above $\delta_{\min}$	0.0	100.0	0.0	0.0
Return present	0.0	100.0	0.0	0.0
High but quantified measurement error	2.5	14.0	83.5	0.0

Scenario	Contradicted	Survived	Inconclusive	Not evaluable
Unreliable recording of B	0.0	0.0	0.0	100.0
Misspecified covariates	0.0	100.0	0.0	0.0
Latent confounding	0.0	100.0	0.0	0.0
Failed positive control	2.0	0.0	0.0	98.0

Table 7. Classification of the conjunction into the four outcomes, as a percentage of 200 simulated studies per scenario. Wilson intervals for a proportion of 100% over 200 studies are [98.1, 100]; for 95.0%, [91.0, 97.3]; for 86.5%, [81.1, 90.6]; for 72.5%, [66.0, 78.2].

Alt text: Heat map with twelve scenarios as rows and four outcomes as columns, each cell labelled with a percentage. Absent return with an adequate sample reads 100 contradicted. Return present, misspecified covariates and latent confounding each read 100 survived. Unreliable recording reads 100 not evaluable and failed positive control reads 98 not evaluable. A vertical red line separates the first three columns from the not-evaluable column.

The gates behave as intended. An unreliable recording of the intermediate population is classified as not evaluable in every study, and a failed positive control in 98.0%.

Under a valid design, an absent return with an adequate sample is contradicted in every study. An effect inside the equivalence zone is contradicted in 95.0% at twenty subjects and in 100% at forty subjects with four hundred trials, so the decision does not depend on the larger sample. An effect at the boundary is predominantly inconclusive (86.5%), which is the intended behaviour at a margin. An effect clearly above the margin, and a present return, survive in every study. High but quantified measurement error, with the components still



Figure 4: Four-outcome classification of the conjunction by scenario. The vertical rule separates decisions taken under a valid design from validity failures.

474 estimable, yields inconclusive in 83.5% and contradiction in 2.5%.

475 The six-subject scenario is informative and is not simply a low-power condition. **The con-**
 476 **junction is still contradicted in 72.5% of those studies.** The reason is structural:
 477 a conjunctive necessary condition is contradicted when *any* required component is contra-
 478 dicted, and under absent return the second component has a null effect that is estimated
 479 precisely even with few subjects, so the decision does not require the sample that a single-
 480 estimand design would demand. Within this simulated design, conjunctive conditions are
 481 therefore harder to satisfy but easier to refute than their individual components. Whether
 482 this extends beyond the generative models simulated here is not established, and the property
 483 should be declared in a preregistration rather than assumed.

484 The two negative scenarios remain central. Under **misspecified covariates** and under
 485 **latent confounding**, the conjunction survives in 100% of studies although no return exists.
 486 No validity gate detects either, because the dependence is real and the recordings are reliable;
 487 what fails is the causal interpretation.

488 4.3 Behaviour with sample size

Conjunction				
positive [95%				
Trials	CI]	$\Delta R^2_{A_1 \rightarrow B}$	$\Delta R^2_{B \rightarrow A_2}$	Chain
250	9.0% [5.8, 13.8]	100.0	51.5	14.0
500	12.5% [8.6, 17.8]	100.0	84.0	14.0
1000	16.0% [11.6,	100.0	97.0	16.0
	21.7]			
2000	11.5% [7.8, 16.7]	100.0	99.5	11.5
4000	18.8% [11.7,	100.0	100.0	18.8
	28.7]			

Table 8. Rate of false return-compatible attribution under one weak-confounding generator with no true return, by number of trials, over 200 replicates per size except the 4000-trial cell, which used 80 replicates and therefore carries a wider interval.

The second component degrades with sample size exactly as expected of a confounded association, rising from 51.5% at 250 trials to 100% at 4000: narrowing the interval of a confounded association increases confidence in an interpretation that was never identified. The conjunction does not follow it. Under this specific weak-confounding generator, the conjunction attenuated false return-compatible attribution across the simulated range of 250 to 4000 trials, remaining between 9.0% and 18.8% with overlapping intervals; increasing the sample size over that range did not increase the chain-level positive rate monotonically.

This is not immunity, and the range examined is finite. The latent-confounder architecture of Table 6 uses a stronger shared driver, and there the chain component reaches 86.5% and the trial-specific latent feedforward state reaches 100%. What the sweep establishes, within the generators and sample sizes simulated, is that the pathology by which accumulating data aggravates a misidentified association operated on segment-wise predictive gains and not on episode identity.

4.4 Sensitivity to the equivalence threshold

Scenario	$\delta_{\min} = 0.005$	$\delta_{\min} = 0.01$	$\delta_{\min} = 0.02$
Absent return, adequate sample	46.0 contradicted, 54.0 inconclusive	100.0 contradicted	100.0 contradicted
Effect within the equivalence zone	28.7 contradicted, 71.3 inconclusive	94.7 contradicted, 5.3 inconclusive	100.0 contradicted
Return present	100.0 survived	100.0 survived	100.0 survived
Latent confounding	100.0 survived	100.0 survived	100.0 survived

Table 9. Classification under three equivalence thresholds, as a percentage of 150 simulated studies per cell. The chain margin was scaled proportionally, $\delta_{\text{epi}} = 2\delta_{\text{min}}$, as was the positive-control recovery requirement.

The threshold modulates the contradiction rate and leaves the central finding untouched. At the strictest margin the procedure becomes conservative, declaring fewer than half of absent-return studies contradicted and the remainder inconclusive, which is the correct response to demanding a smaller effect than the design can resolve. The two scenarios that carry the argument are invariant across the range examined: a present return and latent confounding both yield survival in 100% of studies at all three thresholds. No margin within this range converts the spurious survival under latent confounding into a contradiction, because the confounded dependence is genuinely present and the equivalence test is not the mechanism that would detect its misinterpretation.

4.5 Positive-control diagnostics

Three quantities must be distinguished: the true validity status of a study, the status detected by the algorithm, and the decision produced.

Across 200 studies in which the positive control was broken by trial misalignment of the B recording, 98.0% [95.0, 99.2] were correctly routed to not evaluable. The residual 2.0% [0.7, 5.7] proceeded to an equivalence decision and were classified as contradicted. These are false negatives of the validity diagnostic, not valid contradictions, and are reported as such. Across the 2200 studies in the eleven scenarios with an intact control, no study was routed to not evaluable by the positive-control gate, so its false-positive rate is below the resolution of this simulation; the unreliable-recording scenario was routed by the reliability gate, which precedes it.

A false-negative rate of two per cent in the validity diagnostic is not negligible when the consequence is a spurious contradiction of a theoretical claim. It argues for reporting control-

channel performance alongside every contradiction rather than treating a passed gate as silent.

5. Discussion

5.1 What the example establishes

Stating recurrence topologically and testing it conjunctively yields a criterion substantially more discriminating than early–late dependence. It separates return from local persistence, from feedforward termination, from observed common causes, from parallel pathways, and from coarse population-level dependence, all at or near the nominal rate. It also makes explicit what a bare arrow conceals: that “recurrence” is a claim about where activity goes and about what survives the journey, not merely about when activity is observed.

The token-disruption result establishes something narrower but useful. Predictive gain along each segment of a chain can be entirely real while the episode-specific content that motivated the claim is destroyed in transit. Any criterion built from segment-wise predictive gains alone is blind to this, and the point generalizes beyond the present example to multi-stage informational claims tested by incremental prediction.

A third result concerns how the criterion behaves as data accumulate. The incremental estimand alone shows the pathology familiar from confounded associations, rising from 51.5% to 100% across the simulated range. The conjunction did not, because episode identity was not manufactured by additional observations of a weak shared driver. It was nonetheless defeated by a stronger shared driver at every sample size examined. Accumulating data therefore aggravated misidentification of segment-wise predictive gains but not of chain-level identity, which narrows the problem without solving it.

The example equally establishes the limits. Satisfaction of the full conjunction places a study at Level 1 and no further. Moving to Level 2 requires exogenous variation, selective

perturbation of the intermediate population or of its return pathway, or explicit identification assumptions. Level 3 remains a statement about a necessary condition: even a demonstrated causal return would show only that the condition survived, never that recurrence constitutes presence.

5.2 Implementationist claims

Implementationist theories are a subclass of the claims examined here, and the framework does not adjudicate between computationalism and implementationism. It requires explicit commitments of any position: which implementation property is claimed necessary, how its absence could be established, and what conscious counterexample would contradict the claim. Recurrence functions here only as a worked example, and the article is not a comparison of theories of consciousness.

Consider, as a limiting case, a theory whose biomarker of presence depends on report, whose candidate intervention also affects arousal, whose temporal resolution does not separate the relevant windows, and whose mechanism is defined retrospectively from the result. No available measurement estimates its necessary condition validly, and the correct diagnosis is not evaluable. Naming this outcome prevents both false refutation and false confirmation; the simulation instantiates the corresponding failures directly, classifying an unreliable recording and a failed positive control as not evaluable in 100% and 98.0% of studies respectively.

5.3 Using the framework

A preregistration should state: the variable explained, under a non-circular definition; the mechanism and its scale; the necessary condition, its derivation, and which of the eight requirements it satisfies; if the condition is conjunctive, each component and the aggregation rule; if disjunctive, the disjuncts declared exhaustively in advance; the same-source specification and how it is frozen before analysis; the attribution model φ with its thresholds,

calibration, and inferential independence from M ; the observation model and its error structure; the boundary between D and D_{eval} ; the smallest relevant effect for each component and the analysis supporting it; the conditional null for each component and its exchangeability assumptions; which predictions are core, implementation-level, and auxiliary; and which subsequent modifications count as refinement, implementation change, or core replacement. Exploratory analyses discover candidates; they do not contradict. The framework is most productive when rival theories specify divergent necessary conditions over the same paradigm (Cogitate Consortium et al. 2025).

5.4 Limitations

First, presence continues to depend on a phenomenological characterization and on imperfect inferential bridges. φ disciplines that inference without providing an independent detector of experience, and the underdetermination between constitutive and causal readings has been documented more generally (Michel 2019).

Second, the characterization is synthetic throughout. The generative models are linear or monotone, Gaussian, and homogeneous across subjects, and they do not represent non-stationarity, correlated artefacts, or strong between-subject heterogeneity. The equivalence classification, sample-size analysis, and threshold sensitivity were characterized under the specification reported here, but only across the prespecified synthetic models, parameter ranges, and computational budgets described in the Supplementary Methods.

Third, the validity of the conditional nulls depends on the specification of the conditional model and on the exchangeability of its residuals. Under misspecification the nulls are approximate and may be anticonservative; the misspecified-covariate scenario shows the consequence, not a correction for it.

Fourth, the regularization parameter was held fixed rather than tuned, and the mapping of behaviour across nonlinearities is limited to a single monotone transformation. Power of the

chain component under other nonlinearities is not established.

Fifth, the sample-size and threshold ranges examined are finite, and no behaviour outside 250–4000 trials or outside margins of 0.005–0.02 is characterized.

Sixth, the episodic matching is independent nearest-neighbour assignment rather than a one-to-one assignment. It is therefore insensitive to query ordering, but it does not enforce a bijection between queries and targets, and a global assignment procedure would yield a different, generally more conservative, statistic.

Seventh, application to empirical data remains future work. Trial-resolved intracranial recordings with simultaneous sampling of subcortical and cortical structures during visual consciousness tasks, such as those reported by Fang et al. (2025), would be a natural setting in which to estimate the components, provided that a source population, an intermediate population, two windows, and the same-source mapping can be defined in advance. Under favourable conditions such an analysis could establish compatibility with informational continuity or return; it could not by itself establish causal return, constitution, identity, or the presence of minimal phenomenal experience, and would be further limited by small samples and by paradigms indexing reported visual consciousness.

Eighth, and most consequentially, the observational conjunction does not identify causal return under latent structure. This is a property of the inferential situation rather than of the particular statistics chosen.

Nothing in this article should be read as support for any particular neuroanatomical architecture.

6. Conclusions

What makes a constitutive claim evaluable is deriving a non-trivial necessary physical condition, connecting it to explicit observation and presence-attribution models, separating the

theoretical domain from the domain a given study can evaluate, and accepting in advance both a set of validity gates and a region of failure under a severe test. What such a test permits is exclusion: when a core condition fails reliably within the evaluable domain, with independent evidence of presence, the formulation is excluded.

What it does not permit is confirmation. Survival confirms neither constitution nor identity, and the worked example fixes the limit numerically. Strengthening the observational criterion from a single early–late arrow to a four-part conjunction requiring both segments of the path, chain-level episode identity, and a preregistered source mapping removes five of nine non-return architectures from consideration entirely. It does not remove the two that matter: a trial-specific latent feedforward state satisfies the complete conjunction in every simulated replicate without any return existing, and under the four-outcome rule both latent confounding and misspecified covariates yield survival in every simulated study, at every equivalence threshold examined.

The present results show that this observational criterion cannot substitute for intervention or for additional identification assumptions under the latent structures examined. The broader inference — that observational sharpening will not in general resolve the underdetermination between a topological causal claim and a latent common cause — follows from the structure of the identification problem rather than from these simulations, which can illustrate it but cannot establish it as a theorem. What the simulations do establish is that strengthening the criterion improves discrimination against several latent alternatives — the shared low-rank factor, the token-disrupted chain, and the weak-confounding sweep — but cannot guarantee identification when relevant latent structures remain unmeasured.

Acknowledgements

None.

Reproducibility and data availability

This article generates no new empirical dataset. All reported results derive from a shared core module defining the generative architectures, the segment estimands, the chain-level episodic test, and the conditional-null machinery, together with analysis scripts for each reported table. Random seeds are fixed in each script, and the mapping from script to table and figure is given in the Supplementary Methods.

The article-specific code will be deposited in the public AFH-MODEL repository (<https://github.com/camilosjobergtala/AFH-MODEL>) under the Apache License 2.0 before submission, identified by release tag [INSERT FINAL RELEASE TAG] and commit [INSERT FINAL COMMIT HASH]. An archived record will be available at [INSERT ARCHIVED DOI IF CREATED].

Declarations

Conflicts of interest: none declared.

Funding: this research received no external funding.

AI assistance: artificial-intelligence tools were used for language assistance, critical review, bibliographic verification, and support in implementing and running the simulations. Conceptual and methodological decisions, interpretation of the literature, and final responsibility for the content rest with the author; no authorship or scientific authority is attributed to the tool.

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